



TUSHAR KHARI

Machine Learning & Data Engineer

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Education

Hochschule Aalen – University of Applied Sciences, Germany	2025 – Present
M.Sc. in Machine Learning and Data Analytics	GPA: 1.9 / 5.0 (German Scale)
Panjab University, India	2020 – 2024
B.Eng. in Information Technology	CGPA: 8.47 / 10

Work Experience

Software Engineer	Jan 2024 – Dec 2024
ICS Group, Chandigarh, India	

- Engineered and deployed containerized **FastAPI** microservices via **Docker**, reducing inference response latency to **< 120 ms**.
- Optimized and fine-tuned **PyTorch** deep neural architectures, driving a **17% increase** in multi-class classification accuracy.
- Architected a production **RAG pipeline** using **LangChain** and vector retrieval, automating unstructured document parsing across large internal corpora.
- Overhauled data ingestion by building automated **Python and SQL ETL pipelines**, cutting training data prep time by **30%**.

Projects

Predictive Maintenance System <i>Python, Scikit-learn, XGBoost, FastAPI, Streamlit</i>	Live App
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- Constructed an end-to-end predictive pipeline on the AI4I sensor dataset using **Random Forest** and **XGBoost** classifiers.
- Tuned decision thresholds to **0.75** via precision-recall trade-off analysis, increasing failure recall by **15%** while sustaining high precision.

Modern Data Warehouse & Analytics Pipeline <i>PostgreSQL, PySpark, Databricks, SQL</i>	GitHub
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- Implemented a multi-tier **Medallion Architecture** in **PySpark** to orchestrate ERP and CRM data ingestion into clean analytics layers.
- Formulated star schema dimensional models and indexed analytical queries, eliminating redundant aggregations for dashboard feeds.

Multimodal Sensitive Data Detection <i>PyTorch, YOLO, BLIP-2, Hugging Face, Streamlit</i>	GitHub
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- Combined **YOLO object detection**, **Tesseract OCR**, and zero-shot transformer models into a single inference pipeline for automated PII masking.
- Integrated **BLIP-2** for context-aware scene captioning and deployed an interactive evaluation dashboard in Streamlit.

Technical Skills

Languages & Core: Python, SQL, Java, Bash, Git
Data Engineering: PySpark, Databricks, PostgreSQL, ETL Pipelines, Medallion Architecture, Dimensional Modeling
Machine Learning & Generative AI: PyTorch, Scikit-learn, XGBoost, RAG, LangChain, Transformers, Vector Search, Computer Vision
Deployment & Infrastructure: FastAPI, Docker, CI/CD, Linux, Azure, Streamlit, Power BI

Languages

English: Fluent | **German:** B1 (Working proficiency, progressing toward B2)